Huawei Access Node KR-ACC-72 QoS Congestion Event Resolution SOP

Node: KR-ACC-72 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_6912

Description

This Standard Operating Procedure (SOP) is designed to address a QoS congestion event on Huawei Access Node KR-ACC-72 at KR Puram Access site. QoS congestion events, including microbursts, interface tail drops, and QoS queue starvation, occur due to the interaction between the Access node's hardware and the QoS policy configuration. The Access node's QoS policy is enforced by the Hierarchical Token Bucket (HTB) scheduler, which allocates bandwidth to traffic classes based on their priority and rate settings. When the HTB scheduler is unable to allocate sufficient bandwidth to a high-priority traffic class, it results in QoS queue starvation, leading to microbursts and interface tail drops. This SOP aims to troubleshoot and resolve the root cause of the QoS congestion event on KR-ACC-72 by executing a series of diagnostic commands, identifying the root cause, and implementing corrective actions.

Pre-requisite Checks

- check: Verify the site location and node ID to ensure that the issue is specific to KR-ACC-72 at KR Puram Access site.

- check: Check the redundant link status to ensure that the issue is not related to a link failure or misconfiguration.

- check: Check the power supply and environmental conditions to rule out any external factors contributing to the issue.

Potential Root Causes

Cause	Probability
QoS policy misconfiguration	High
HTB scheduler misconfiguration	Medium
Hardware resource shortage	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy, Expected Output: QoS policy configuration details, Description: This command displays the QoS policy configuration, including the traffic classes, priority, and rate settings. It helps identify any potential misconfiguration or inconsistencies in the QoS policy.

Command: display qos queue, Expected Output: QoS queue statistics, Description: This command displays the QoS queue statistics, including the number of packets dropped, the number of packets transmitted, and
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the queue utilization. It helps identify any queue starvation or
congestion issues.

Command: display interface, Expected Output: Interface statistics,
Description: This command displays the interface statistics, including
the number of packets transmitted, the number of packets received, and
the interface utilization. It helps identify any potential issues with
the interface or link.
```

Resolution Steps

Step 1: Step: Verify and Correct QoS Policy Configuration, Command: undo qos policy [traffic-class-name], Description: If the QoS policy configuration is found to be misconfigured, this command is used to undo the configuration. The corrected configuration should be applied using the "qos policy" command.

Step 2: Step: Verify and Correct HTB Scheduler Configuration, Command: undo qos queue [queue-name], Description: If the HTB scheduler configuration is found to be misconfigured, this command is used to undo the configuration. The corrected configuration should be applied using the "qos queue" command.

Step 3: Step: Verify and Correct Hardware Resource Allocation, Command: undo qos resource [resource-name], Description: If the hardware resource allocation is found to be misconfigured, this command is used to undo the configuration. The corrected configuration should be applied using the "qos resource" command.

Post-Check Verification

```
Command: display qos policy, Expected Output: QoS policy configuration
details, Description: This command is used to verify that the QoS
policy configuration has been corrected.

Command: display qos queue, Expected Output: QoS queue statistics,
Description: This command is used to verify that the QoS queue
statistics have improved.

Command: display interface, Expected Output: Interface statistics,
Description: This command is used to verify that the interface
statistics have improved.
```

Roll-back Procedures

- Step: Restore Original QoS Policy Configuration, Command: undo undo qos policy [traffic-class-name], Description: If the correction steps fail to resolve the issue, this command is used to restore the original QoS policy configuration.

- Step: Restore Original HTB Scheduler Configuration, Command: undo undo qos queue [queue-name], Description: If the correction steps fail to resolve the issue, this command is used to restore the original HTB scheduler configuration.

- Step: Restore Original Hardware Resource Allocation, Command: undo undo qos resource [resource-name], Description: If the correction steps fail to resolve the issue, this command is used to restore the original hardware resource allocation.